

Discussion on “Reoptimization in the dynamic and stochastic knapsack problem: from near-optimality to distributional equivalence” by Arlotto, Kuo and Xie

Discussant: Velibor V. Mišić

UCLA Anderson School of Management

Advances in Decision Analysis Conference, June 2026

Dynamic stochastic knapsack problem

- ▶ We have a knapsack with capacity c
- ▶ We have a finite horizon of T periods
- ▶ Each period, we are shown an object with weight W_t , and we can choose to accept or reject
 - ▶ Accepting consumes W_t units of capacity, and gives us 1 unit of reward
 - ▶ Rejecting consumes no capacity and yields no reward
- ▶ The weights $W_1, \dots, W_T$ are drawn IID from a known distribution F
- ▶ How do we decide dynamically whether to accept or to reject an item?

Adaptive policy

- ▶ The adaptive policy is motivated using the notion of a consumption function
- ▶ The paper briefly mentions that the consumption function arises in the optimal solution of the relaxation of the DSKP
- ▶ I wanted to informally show this

Formulating the problem as an LP

Formulating the problem as an LP

- ▶ Suppose that F has finite support: $w_1, \dots, w_S$, with probabilities $p_1, \dots, p_S$

Formulating the problem as an LP

- ▶ Suppose that F has finite support: $w_1, \dots, w_S$, with probabilities $p_1, \dots, p_S$
- ▶ We can then write the following deterministic linear program:

$$\begin{aligned} & \text{maximize} && \sum_{s=1}^S x_s \\ & \text{subject to} && \sum_{s=1}^S w_s x_s \leq c \\ & && 0 \leq x_s \leq T p_s, \quad \forall s \in [S]. \end{aligned}$$

Formulating the problem as an LP

Formulating the problem as an LP

- ▶ Perform a change of variables: define y_s where $x_s = p_s \cdot y_s$

Formulating the problem as an LP

- ▶ Perform a change of variables: define y_s where $x_s = p_s \cdot y_s$
- ▶ Then problem becomes

$$\text{maximize} \quad \sum_{s=1}^S p_s \cdot y_s \quad (1)$$

$$\text{subject to} \quad \sum_{s=1}^S w_s \cdot p_s \cdot y_s \leq c \quad (2)$$

$$0 \leq y_s \leq T, \quad \forall s \in [S]. \quad (3)$$

Formulating the problem as an LP

- ▶ Perform a change of variables: define y_s where $x_s = p_s \cdot y_s$
- ▶ Then problem becomes

$$\text{maximize} \quad \sum_{s=1}^S p_s \cdot y_s \quad (1)$$

$$\text{subject to} \quad \sum_{s=1}^S w_s \cdot p_s \cdot y_s \leq c \quad (2)$$

$$0 \leq y_s \leq T, \quad \forall s \in [S]. \quad (3)$$

- ▶ Optimal solution is greedy where all but possibly one s are set to either 0 or T

Formulating the problem as a (semi-infinite) LP

Formulating the problem as a (semi-infinite) LP

- ▶ Suppose we now allow F to have infinite support

Formulating the problem as a (semi-infinite) LP

- ▶ Suppose we now allow F to have infinite support
- ▶ Then problem should be

$$\begin{aligned} &\text{maximize} && \int_0^\infty y(w) dF(w) \\ &\text{subject to} && \int_0^\infty y(w) \cdot w dF(w) \leq c \\ &&& 0 \leq y(w) \leq T, \quad \forall w \in [0, \infty) \end{aligned}$$

Formulating the problem as a (semi-infinite) LP

- ▶ Suppose we now allow F to have infinite support
- ▶ Then problem should be

$$\begin{aligned} & \text{maximize} && \int_0^\infty y(w) dF(w) \\ & \text{subject to} && \int_0^\infty y(w) \cdot w dF(w) \leq c \\ & && 0 \leq y(w) \leq T, \quad \forall w \in [0, \infty) \end{aligned}$$

- ▶ Optimal solution (should be) extremal (each $y(w) \in \{0, T\}$ for all w)
- ▶ And just like finite knapsack problem, it is greedy, i.e., there should be threshold ϵ such that

$$y(w) = \begin{cases} T & \text{if } w \leq \epsilon, \\ 0 & \text{if } w > \epsilon \end{cases}$$

Optimal objective value is then $TF(\epsilon)$

Deriving the consumption function

Deriving the consumption function

- What should ϵ be? It should be such that main constraint is tight, i.e.,

$$\int_0^{\infty} y(w) \cdot w dF(w) = c$$

Deriving the consumption function

- What should ϵ be? It should be such that main constraint is tight, i.e.,

$$\int_0^{\infty} y(w) \cdot w dF(w) = c$$

- Using structure of optimal y , this means

$$\int_0^{\epsilon} Tw dF(w) = c$$

or equivalently

$$\int_0^{\epsilon} w dF(w) = \frac{c}{T}$$

Deriving the consumption function

- ▶ What should ϵ be? It should be such that main constraint is tight, i.e.,

$$\int_0^{\infty} y(w) \cdot w dF(w) = c$$

- ▶ Using structure of optimal y , this means

$$\int_0^{\epsilon} Tw dF(w) = c$$

or equivalently

$$\int_0^{\epsilon} w dF(w) = \frac{c}{T}$$

- ▶ With k periods remaining, then ϵ is the threshold such that $\int_0^{\epsilon} w dF(w) = c/k$

Deriving the consumption function

- What should ϵ be? It should be such that main constraint is tight, i.e.,

$$\int_0^{\infty} y(w) \cdot w dF(w) = c$$

- Using structure of optimal y , this means

$$\int_0^{\epsilon} Tw dF(w) = c$$

or equivalently

$$\int_0^{\epsilon} w dF(w) = \frac{c}{T}$$

- With k periods remaining, then ϵ is the threshold such that $\int_0^{\epsilon} w dF(w) = c/k$
- This is precisely the consumption function (modulo some handwaving)

Key message of the paper

- ▶ Two different policies for the dynamic and stochastic knapsack problem (DSKP) that are asymptotically optimal:
 - ▶ Non-adaptive: accept if W_t is below a static threshold (constant over time)
 - ▶ Adaptive: accept if W_t is below a dynamic threshold
- ▶ Non-adaptive, adaptive and DP policy have the same asymptotic mean performance ...
- ▶ ... but a different variance
 - ▶ In the unit capacity, standard uniform setting: adaptive reduces variance by $\sim 27\%$ over non-adaptive
- ▶ Paper shows that adaptive and non-adaptive reward distributions are asymptotically normal

Why is this interesting

- ▶ Virtually all research in dynamic programming focuses on expected reward
 - ▶ We solve this multi-stage decision making problem over and over, and we care about performance in the aggregate
- ▶ Less common to consider the variance
- ▶ Other work that considers uncertainty / risk in Markov decision processes:
 - ▶ Iyengar (2005), Nilim and El Ghaoui (2005), Mannor, Simester, Sun and Tsitsiklis (2007), Delage and Mannor (2010), Wiesemann, Kuhn and Rustem (2013), Goyal and Grand-Clement (2023), ...

Some thoughts that come to mind

- ▶ The main result of this paper suggests that we can look at dynamic problems (such as DSKP) as multi-objective problems
- ▶ **Question:** what does the trade-off between expected reward and the variance of the reward look like?
 - ▶ Can a small reduction in mean performance greatly reduce variance?

Trading-off mean reward and variance

- ▶ To fix things, let's consider the unit capacity, standard uniform setup (i.e., $c = 1$, $W_t \sim \text{Uniform}(0, 1)$)
 - ▶ Adaptive policy: accept if $W_t \leq \min\{x_t, \sqrt{2x/(T - t + 1)}\}$
 - ▶ Non-adaptive policy: accept if $W_t \leq \min\{x_t, \sqrt{2c/T}\}$
- ▶ We will consider a couple of “dumb” benchmarks for studying this trade-off

A benchmark to trade-off mean and variance

► Benchmark 1: a 2-sided rule:

- Fix $q \in (0, 1)$
- Two-sided non-adaptive policy: accept item at t iff

$$q \cdot \sqrt{2/T} \leq W_t \leq \min\{x, \sqrt{2/T}\}$$

- Two-sided adaptive policy: accept item at t iff

$$q \cdot \sqrt{2x/(T - t + 1)} \leq W_t \leq \min\{x, \sqrt{2x/(T - t + 1)}\}$$

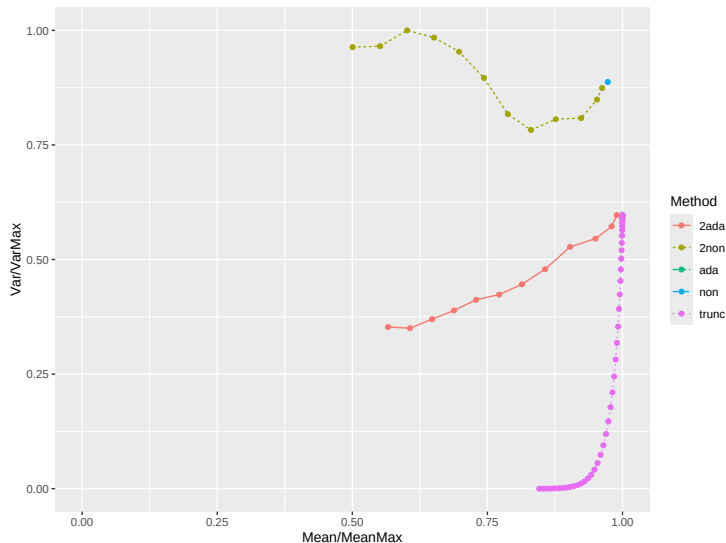
Another benchmark to trade-off mean and variance

► Benchmark 2:

- Set a target τ
- Run adaptive policy as normal, but **stop** once the number of accepted items is τ
- Performance will look like $\min\{R(\pi), \tau\}$, which is (asymptotically) a truncated normal distribution

Some results

$(c = 1, T = 1000, W_t \sim \text{Uniform}(0, 1).)$



Other thoughts - I

- ▶ Is there a more profitable trade-off than the truncated policy (purple curve)?
- ▶ Do there exist asymptotically optimal policies with even lower variance than adaptive policy?

Other thoughts - II

- ▶ What does the mean-variance story look like for other dynamic stochastic problems?
 - ▶ Problem is similar to network revenue management
 - ▶ Policy somewhat resembles probabilistic allocation control (Jasin and Kumar 2012)
 - ▶ Can analysis be ported to this setting?

Other thoughts - III

- ▶ Is there a hidden robustness story?
 - ▶ Gotoh, Kim and Lim (2018): robust empirical optimization is almost the same as mean-variance optimization
 - ▶ Provides an explanation for why robust optimization performs well out-of-sample: robustness reduces variability of reward distribution
 - ▶ When robustness parameter is small, reduction in variance is order of magnitude larger than reduction in expected value
 - ▶ Suppose we design adaptive policy (thresholds) based on F , but actual distribution is F' ; does adaptive fare better than non-adaptive?

Thank you!

Extra material